

From a witness of quantumness to a bosonic code and back to the hardware: the spider state, rotation-symmetric codes, and a drive-tuned CROT gate

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Rotation-symmetric bosonic codes encode a qubit in the Fock states $|mK\rangle$ of a single oscillator. We follow a primitive that was found for an unrelated reason: the SPIDER state, which maximally violates the classical bound of the precession protocol of Zaw *et al.* [Phys. Rev. A **106**, 032222 (2022)] and happens to have exactly the required K -fold rotation symmetry. We show that the score operator of the protocol is block diagonal in the rotation-symmetry sectors, so that the spider state defines an order- K code whose preparation can be certified device-independently from sign measurements alone, with a spectral-gap fidelity witness. The code, however, is not error-correcting: its codewords carry mean photon numbers in a fixed ratio of three, the Knill–Laflamme non-deformation condition fails by an amount of order the photon number, no finite reweighting of the coefficients can repair it without abandoning the maximal violation that defines the state, and the exact state has infinite mean energy. Since the obstacle to rotation-code error correction is then not the primitive but the gadgets, we turn to the controlled-rotation gate $e^{i\varphi\hat{n}\otimes\hat{n}}$ that every rotation code needs and engineer it from the drive-tuned cross-Kerr interaction between two superconducting cavities sharing a transmon. An untargeted drive enhances the cross-Kerr four-fold and yields a gate time of tens of microseconds, but produces self-Kerr of comparable size; a 99% gate tolerates a residual self-Kerr below one percent of the cross-Kerr. We then reproduce, with the same dressed-ladder numerics, the single-tone drive-enhanced scheme that completed the project [Qin *et al.*, arXiv:2609.07076]: a drive placed next to the two-cavity conversion resonance raises the cross-Kerr on-off ratio above fifty while the self-Kerrs stay small, and a driven side transmon—already present in recovered four-mode simulations—cancels the residual, leaving the data rail of the CROT gate exact in the diagonal model.

I. INTRODUCTION

Bosonic quantum error correction (QEC) replaces the many physical qubits of a conventional code by the many levels of a single oscillator [1–3]. The redundancy is free, the dominant error (photon loss) is simple, and the hardware—three-dimensional superconducting cavities read out by a transmon ancilla—is mature enough that a cat code [4] and more recently a GKP code [5] have been operated beyond the break-even point. Among single-mode codes, the family defined by a discrete rotation symmetry in phase space, the *rotation-symmetric bosonic* (RSB) codes of [6], is particularly attractive: it contains the cat [7] and binomial [8] codes, its distance is set by a single integer K , and it comes with a complete gate set and a teleportation-based error-correction gadget built from two primitives, a controlled-rotation gate $\text{CROT}(\varphi) = e^{i\varphi\hat{n}\otimes\hat{n}}$ and a phase measurement.

The physics of RSB codes is, to a large extent, the physics of the *primitive state* from which the codewords are built. A coherent state gives a cat code, a binomial amplitude profile gives a binomial code, a phase state gives the number-phase code. It is therefore natural to ask whether other states with K -fold rotation symmetry—states found for entirely different reasons—make interesting codes. The present work follows one such candidate from its discovery to its verdict, and then to the hardware question that the verdict leaves standing.

The candidate comes from the foundations of quantum

mechanics. Tsirelson [9] and, in a general form Zaw *et al.* [10] showed that a harmonic oscillator in uniform precession obeys a classical bound on the probability of finding its coordinate positive when sampled at K equally spaced times, and that this bound is violated by quantum states. The state that violates it *maximally* has support only on the Fock states $|mK\rangle$, and its Wigner function has K legs of negativity emanating from the origin: the SPIDER state. Because support on $|mK\rangle$ is precisely the definition of a logical $|+\rangle$ of an order- K RSB code, the spider state is a code in waiting, and a special one: it is the only RSB primitive whose preparation can be certified from sign measurements alone.

We ask three questions and answer each numerically with a small, open code base. (i) What are the properties of the spider state as a code primitive? (ii) Does the resulting spider code satisfy the Knill–Laflamme conditions [11]? The answer is no, and we show why no finite repair of the coefficients can change that. (iii) Since the obstacle is not the primitive but the gadgets, how good a CROT gate can be engineered between two cavities with the drive-tuned cross-Kerr interaction of Zhang *et al.* [12]? For the last question we present our generic exploration and then the answer that the same group has since published, the single-tone drive-enhanced CROT gate of Qin *et al.* [13], which we reproduce with the same numerical tools. The paper is organised along these questions: Sec. II introduces the precession protocol and the spider state, Sec. III builds the spider code, Sec. IV delivers the Knill–Laflamme verdict, and Sec. V turns to

the cross-Kerr CROT gate. Section VI concludes.

II. THE SPIDER STATE

A. Precession protocol and classical bound

Consider an oscillator $H = \frac{1}{2}(p^2 + x^2)$ in units $\hbar = \omega = 1$, so that $x(t)$ precesses uniformly with period $T = 2\pi$. In each run of the protocol the state is prepared, one of K times $t_k = kT/K$ ($k = 0, \dots, K-1$) is drawn at random, and the sign of x is recorded. The score is the average probability of a positive sign,

$$P_K = \frac{1}{K} \sum_{k=0}^{K-1} \Pr[x(t_k) > 0]. \quad (1)$$

For *any* classical state of the oscillator—any probability distribution on phase space, however non-Gaussian—one has for odd K [9, 10]

$$P_K \leq \mathbf{P}_K^c = \frac{1}{2} \left(1 + \frac{1}{K}\right), \quad (2)$$

while for even K the score is exactly $1/2$. The bound follows from the observation that a classical trajectory visits the K measurement angles in turn, so at most $\lceil K/2 \rceil$ of them can lie in the half-plane $x > 0$.

Quantum mechanically $P_K = \langle \psi | \hat{Q}_K | \psi \rangle$ with the score operator

$$\hat{Q}_K = \frac{1}{K} \sum_{k=0}^{K-1} \hat{U}_k^\dagger \text{pos}(\hat{x}) \hat{U}_k, \quad \hat{U}_k = e^{-i\hat{n}t_k}, \quad (3)$$

where $\text{pos}(\hat{x}) = \frac{1}{2}(1 + \text{sgn } \hat{x})$ projects on $x > 0$. The largest eigenvalue $\mathbf{P}_K^\infty$ of $\hat{Q}_K$ exceeds $\mathbf{P}_K^c$; its eigenvector is the spider state $|S_K\rangle$.

B. Rotation symmetry from the Kronecker comb

The structure that connects the protocol to bosonic codes is elementary. Since $\hat{U}_k|n\rangle = e^{-2\pi i n k/K}|n\rangle$,

$$(\hat{Q}_K)_{mn} = \text{pos}_{mn} \frac{1}{K} \sum_{k=0}^{K-1} e^{2\pi i(m-n)k/K} = \text{pos}_{mn} \delta_{m \equiv n \pmod{K}} \quad (4)$$

The score operator is the projector $\text{pos}(\hat{x})$ with every matrix element between Fock states of different residue modulo K erased (Fig. 1). It is block diagonal in the residue classes, its eigenvectors can be chosen inside one class, and the top eigenvector lives in the class $n \equiv 0$: it has support only on $|mK\rangle$ and is therefore an eigenstate of the discrete rotation $\hat{R}_K = e^{2\pi i \hat{n}/K}$ with eigenvalue $+1$. The matrix elements of $\text{pos}(\hat{x})$ themselves have the closed form (Appendix A)

$$\langle m | \text{pos}(\hat{x}) | n \rangle = \frac{\psi_n(0)\psi'_m(0) - \psi_m(0)\psi'_n(0)}{2(m-n)} \quad (m \neq n), \quad (5)$$

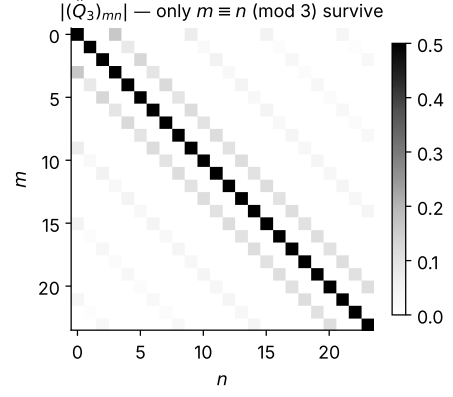


FIG. 1. Modulus of the matrix elements of the score operator $\hat{Q}_3$ in the Fock basis. Only elements with $m \equiv n \pmod{3}$ survive the sum over measurement times, and within a residue class only $m - n$ odd multiples of 3 are non-zero (parity of $\text{pos}(\hat{x})$).

with $\langle n | \text{pos}(\hat{x}) | n \rangle = \frac{1}{2}$, so that truncations of 10^4 Fock states are diagonalised in seconds.

C. Properties of the spider state

Figure 2 summarises the numerics for $K = 3$ (notebook 01 of the repository). The Fock-grid coefficients $s_m = \langle mK | S_K \rangle$ are real with the sign pattern $(-1)^{\lceil m/2 \rceil}$, $s_0 = 0.699$, $s_1 = -0.515$, $s_3 = 0.257$, in agreement with Ref. [10]. The score converges with the truncation n of the Fock grid (cutoff Kn) as a power law, $P_3(n) = \mathbf{P}_3^\infty - 0.040 n^{-0.50}$, to

$$\mathbf{P}_3^\infty = 0.7094 > \mathbf{P}_3^c = 0.6667, \quad (6)$$

and the K measurement times contribute equally, as rotation symmetry requires. For $K = 5, 7, 9$ we find $\mathbf{P}_K^\infty = 0.640, 0.608, 0.589$ against $\mathbf{P}_K^c = 0.600, 0.571, 0.556$, with spectral gaps $\epsilon = q_d - q_{d-1}$ of 0.040 – 0.034 to the next eigenvalue.

The same figure shows the first warning sign. The mean photon number of the truncated spider state grows as $\sqrt{n}$ and never saturates: the coefficient tail decays so slowly that $\sum_m |s_m|^2$ converges but $\sum_m m |s_m|^2$ does not. The exact spider state is normalisable but has *infinite mean energy*. Any physical preparation is a truncated approximation, and its “size” is a free parameter rather than a property of the code. The Wigner functions (Fig. 3) show the K legs of concentrated negativity that give the state its name.

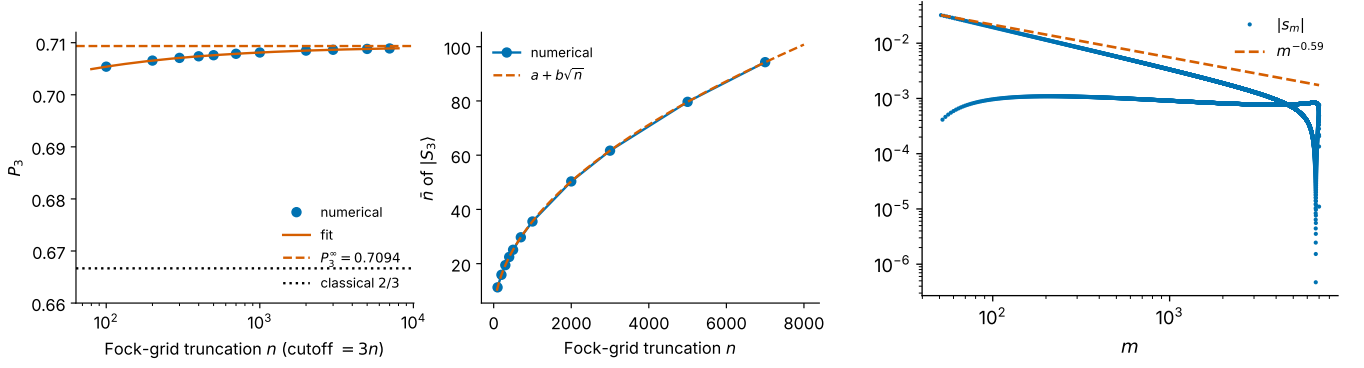


FIG. 2. Left: score of the spider state $|S_3\rangle$ against the Fock-grid truncation n (dots), power-law fit (solid) and extrapolated $P_3^\infty = 0.7094$ (dashed); the classical bound is dotted. Middle: mean photon number of the truncated spider state, which grows as $\bar{n} \simeq 1.13\sqrt{n}$ without saturating. Right: tail of the Fock-grid coefficients $|s_m|$ on a log-log scale; the decay is a power law with exponent between $-1/2$ and $-3/4$, i.e. $|s_m|^2$ decays slower than $m^{-3/2}$.

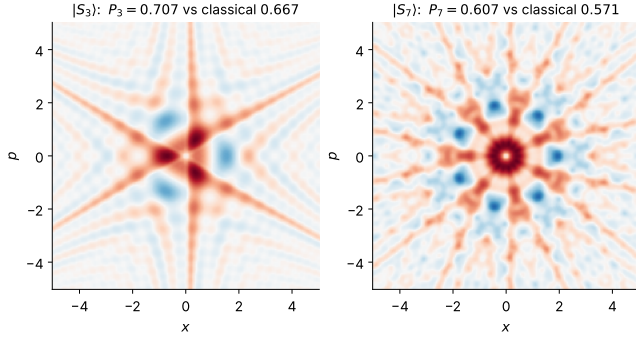


FIG. 3. Wigner functions of the spider states for $K = 3$ and $K = 7$ (truncated to 150 Fock states). Red is positive, blue negative.

III. THE SPIDER CODE

A. Construction

An order- K RSB code [6] is a two-dimensional subspace on which $\hat{Z}_L = e^{i\pi\hat{n}/K}$ acts as the logical Z . The codewords are thus supported on interleaved Fock grids,

$$|0_L\rangle = \sum_m c_{2mK} |2mK\rangle, \quad |1_L\rangle = \sum_m c_{(2m+1)K} |(2m+1)K\rangle, \quad (7)$$

and the dual codewords $|\pm_L\rangle$ have support on every $|mK\rangle$. Any normalised state on the full K -grid can be declared to be $|+_L\rangle$; the Kronecker-comb projectors $\hat{\Pi}_{2K}^\ell = \frac{1}{2K} \sum_{j=0}^{2K-1} (e^{-i\pi\ell/K} \hat{Z}_L)^j$ with $\ell = 0, K$ then extract $|0_L\rangle$ and $|1_L\rangle$. We define the spider code by $|+_L\rangle \equiv |S_K\rangle$. Numerically the codewords satisfy $\hat{Z}_L|0_L\rangle = |0_L\rangle$, $\hat{Z}_L|1_L\rangle = -|1_L\rangle$ and $\hat{R}_K|\pm_L\rangle = |\pm_L\rangle$ to machine precision (Fig. 4).

B. Figures of merit

The codewords are strikingly unequal. $|0_L\rangle$ is dominated by the vacuum and $|1_L\rangle$ by $|3\rangle$, and the mean photon numbers of the truncated codewords are in the ratio $\bar{n}_1/\bar{n}_0 \rightarrow 3.00$ at every truncation (Table I), while both diverge. The embedded phase uncertainty $\Delta_K(\theta) = |\langle e^{iK\theta} \rangle|^{-2} - 1$, which controls how well a code state serves as a phase reference in the error-correction gadget [6], is $\Delta_3 \simeq 3.6\text{--}3.9$ for the spider state, an order of magnitude larger than for cat, binomial or Pegg–Barnett primitives of the same $\bar{n}$ (Fig. 5): the energy in the long Fock tail does nothing for the phase resolution.

C. What the spider code has that no other code has

The one genuinely new feature is certification. Cat and binomial codewords are verified by Wigner tomography. The spider state is by definition the top eigenvector of an observable, $\hat{Q}_K$, that is measured by sign measurements of x at K times—the simplest measurement an oscillator admits. For any state ρ , $\text{tr}(\rho\hat{Q}_K) \leq q_d F + q_{d-1}(1 - F)$

TABLE I. Mean photon numbers of the spider codewords for $K = 3$ as a function of the Fock-grid truncation n (cutoff $3n$).

n	$\bar{n}(0_L)$	$\bar{n}(1_L)$	$\bar{n}_{\text{code}}$	$\bar{n}_1/\bar{n}_0$
100	5.74	16.77	11.26	2.92
300	9.77	29.13	19.45	2.98
1000	17.78	53.32	35.55	3.00
3000	30.81	92.52	61.67	3.00
7000	47.12	141.48	94.30	3.00

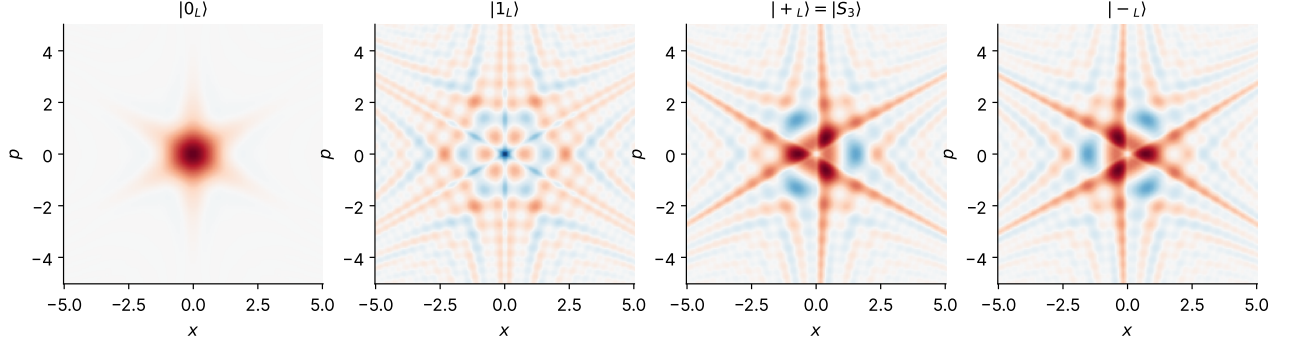


FIG. 4. Wigner functions of the spider-code codewords for $K = 3$. $|0_L\rangle$ is 98% vacuum; $|1_L\rangle$ has 53% of its weight on $|3\rangle$; $|-_L\rangle$ is $|+_L\rangle = |S_3\rangle$ rotated by $\pi/3$.

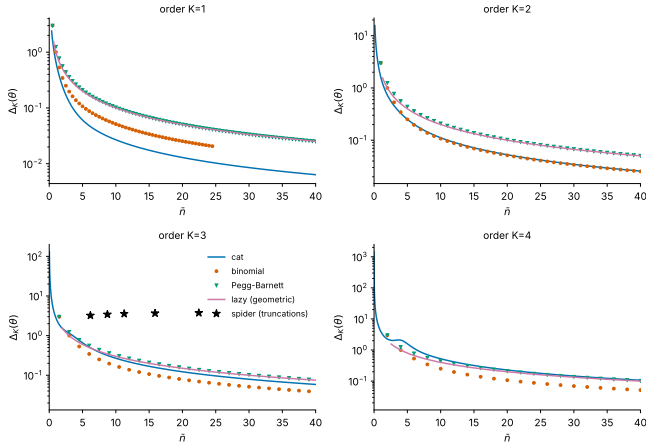


FIG. 5. Embedded phase uncertainty against mean photon number for cat, binomial, Pegg-Barnett and geometric (“lazy”) primitives of order $K = 1$ –4, and for truncations of the spider state (stars; $K = 3$ only, the even- K and $K = 1$ protocols have no non-trivial optimum).

with $F = \langle S_K | \rho | S_K \rangle$, hence

$$F \geq \frac{\text{tr}(\rho \hat{Q}_K) - q_{d-1}}{q_d - q_{d-1}} = 1 - \frac{q_d - \text{tr}(\rho \hat{Q}_K)}{\epsilon}. \quad (8)$$

A measured score within 0.01 of the maximum certifies a fidelity of at least 0.76 with $|S_3\rangle$, device-independently. (The original manuscript in the repository sketched this bound in a weaker form.)

IV. KNILL-LAFLAMME VERDICT

A code corrects the error set $\{\hat{E}_a\}$ if and only if [11, 14]

$$\langle i_L | \hat{E}_a^\dagger \hat{E}_b | j_L \rangle = C_{ab} \delta_{ij}. \quad (9)$$

For $i \neq j$ this is orthogonality: no error maps one codeword onto the other. For $i = j$ it is non-deformation:

TABLE II. Knill–Laflamme matrix elements for order-3 codes with the error set $\{\hat{1}, \hat{a}, \hat{a}^2, \hat{n}\}$ (truncation 600). “Orth.” is the largest $|\langle 0_L | \hat{E}_a^\dagger \hat{E}_b | 1_L \rangle|$; the deformation columns give $\langle 1_L | \hat{E}^\dagger \hat{E} | 1_L \rangle - \langle 0_L | \hat{E}^\dagger \hat{E} | 0_L \rangle$.

code	orth.	$\hat{a}$	$\hat{a}^2$	$\hat{n}$
spider, $K = 3$	0	15.75	247	262
cat, $\alpha = 2$	0	−1.76	−15.9	−17.6
cat, $\alpha = 3$	0	0.36	3.43	3.79
binomial, $L = 2$	0	0	−9.0	−9.0
binomial, $L = 4$	0	0	0	0

both codewords must see the same error statistics, otherwise the syndrome measurement learns, and therefore destroys, the logical information. For an RSB code and loss errors $\hat{a}^l$ with $l < K$, orthogonality is automatic from the Fock-grid spacing. Non-deformation for a single loss reads $\langle 0_L | \hat{n} | 0_L \rangle = \langle 1_L | \hat{n} | 1_L \rangle$: the two codewords must have the same mean photon number.

Table II shows the verdict. The spider code passes orthogonality exactly and fails non-deformation already for a single loss, by an amount of order $\bar{n}$ itself; the relative mismatch $(\bar{n}_1 - \bar{n}_0)/(\bar{n}_1 + \bar{n}_0)$ locks at 0.50 for every truncation (Fig. 6, left inset of notebook 03). By contrast the binomial code with $L = 4$ satisfies all conditions exactly, and the cat code approaches them exponentially in $|\alpha|^2$. The spider code is an error-*detecting* but not an error-*correcting* code.

Could the coefficients be “renormalised” to repair this? Figure 6 answers with the simplest one-parameter family. One parameter fixes one condition: at $p \simeq 0.6$ the single-loss deformation vanishes, but the double-loss deformation remains in the thousands, and each further error would demand a further parameter—the family converges to a binomial-like amplitude profile, not to the spider. More importantly, the repaired $|+_L\rangle$ is no longer the top eigenvector of $\hat{Q}_K$: its score drops from 0.7066 to 0.6987 (still above the classical bound, that part is robust), so the certificate of Eq. (8) is lost. And no

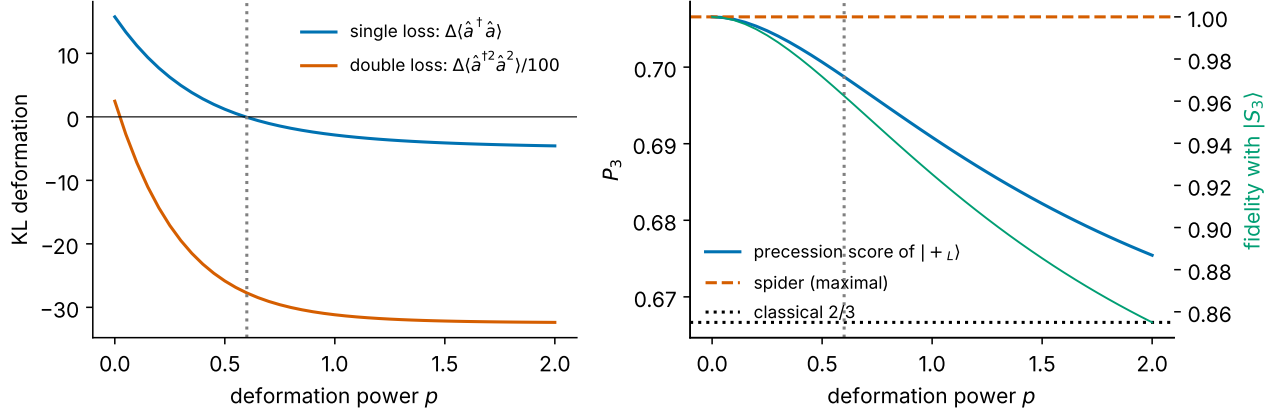


FIG. 6. Attempting to repair the spider code by reweighting $|1_L\rangle \rightarrow \mathcal{N}(\hat{n}+1)^{-p/2}|1_L\rangle$. Left: the Knill–Laflamme deformation for one and two losses as a function of p ; the single-loss deformation vanishes at $p \simeq 0.6$ while the double-loss one is in the thousands. Right: precession score (blue) and fidelity with $|S_3\rangle$ (green) of the resulting $|+_L\rangle$; the repaired state is no longer the spider state.

reweighting of finitely many coefficients touches the infinite energy. Maximal violation of the precession bound and non-deformation under loss are incompatible demands on the same set of Fock-grid coefficients.

V. THE CROT GATE FROM A DRIVE-TUNED CROSS-KERR INTERACTION

A. Why CROT

If the exotic primitive does not help, the limiting factor of RSB error correction is not the primitive but the gadgets, which are the same for every primitive. The hybrid Steane–Knill circuit of Ref. [6] (Fig. 7) couples the data mode (order K_A) to an ancilla (order K_B) by

$$\text{CROT}(\varphi) = e^{i\varphi \hat{n}_A \otimes \hat{n}_B}, \quad \varphi = \frac{2\pi}{K_A K_B}, \quad (10)$$

which acts as the identity on the code space and converts a loss of k photons in the data mode into a rotation by $-2\pi k/(K_A K_B)$ of the ancilla, read out by a phase measurement. The natural generator of CROT is the cross-Kerr interaction $\chi_{AB} \hat{n}_A \hat{n}_B$ between two cavities, which they inherit when both are dispersively coupled to a common transmon [15, 16]. Zhang *et al.* [12] showed that an off-resonant drive on the transmon tunes the inherited self- and cross-Kerr, including through zero.

B. Model

We take a transmon with $\omega_{10}/2\pi = 4.936$ GHz and anharmonicity $\alpha/2\pi = 168$ MHz in the Duffing approximation, capacitively coupled to cavities A and B and

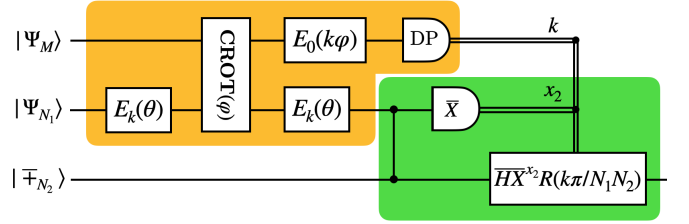


FIG. 7. Hybrid Steane–Knill error-correction gadget for rotation codes: a CROT gate between the data mode $|\Psi_{N_1}\rangle$ and an ancilla $|\Psi_M\rangle$, a phase measurement (DP) and an $\bar{X}$ measurement, followed by a classically controlled recovery on a fresh mode. Reproduced from Grimsmo *et al.*, Phys. Rev. X **10**, 011058 (2020), CC-BY 4.0.

driven at ω_d . In the drive frame and after the rotating-wave approximation [12],

$$\begin{aligned} \hat{H} = & -\delta_{da}\hat{a}^\dagger\hat{a} - \delta_{db}\hat{b}^\dagger\hat{b} - \delta_{dc}\hat{c}^\dagger\hat{c} - \frac{\alpha}{2}\hat{c}^\dagger\hat{c}(\hat{c}^\dagger\hat{c}+1) \\ & + \Omega_d(\hat{c} + \hat{c}^\dagger) + g_a(\hat{a}\hat{c}^\dagger + \text{h.c.}) + g_b(\hat{b}\hat{c}^\dagger + \text{h.c.}), \end{aligned} \quad (11)$$

with $\delta_{dx} = \omega_d - \omega_x$. The dimensionless knobs are $r_1 = \delta_a/\alpha$, $r_2 = g_a/\delta_a \ll 1$ (dispersive regime), $r_3 = \delta_d/\alpha$ and the drive amplitude $r_4 = \Omega_d/\delta_d$. The cavity nonlinearities are read from the dressed energy ladder $E(N_A, N_B)$ of the eigenstates adiabatically connected to $|N_A, N_B\rangle \otimes |\text{transmon ground}\rangle$, obtained by exact diagonalisation and maximum-overlap labelling:

$$E(N) = E_0 + \delta N + \frac{\kappa}{2}N^2 + \frac{\beta}{6}N^3 + \dots, \quad (12)$$

$$\chi_{AB} = E(1, 1) - E(1, 0) - E(0, 1) + E(0, 0). \quad (13)$$

This dressed-ladder method reproduces the time-domain “pseudo-experimental” extraction of the original notebooks (fitting Wigner-function evolutions) a thousand times faster and converges to four digits at modest truncations.

C. Results

For the baseline $r_1 = 9.64$, $r_2 = 0.064$, $r_3 = 0.08$ and no drive, the inherited self-Kerr is $K_A = -2.62$ kHz. Under the drive it changes sign, passing through a Kerr-free point near $|\Omega_d/\delta_d|^2 \simeq 0.05$, and reaches +14 kHz (Fig. 8, left), in agreement with Ref. [12]. With a second cavity on the other side of the transmon ($r_{1b} = -9$, $r_{2b} = 0.05$, $r_3 = 0.26$) the cross-Kerr grows from -1.40 kHz to +5.75 kHz, a four-fold enhancement in magnitude (Fig. 8, right). But K_A , K_B and χ_{AB} all cross zero at nearby yet different drive amplitudes, and everywhere $|K_A|, |K_B| \sim |\chi_{AB}|$: one transmon with one drive can null one self-Kerr, not both, and not at the maximum of χ_{AB} .

The consequence for the gate is severe. We target $\varphi = 2\pi/9$ between an order-3 binomial data code ($L = 4$, the Knill–Laflamme-correct code of Table II) and an order-3 cat ancilla, generated by $\hat{H}_{\text{eff}} = \chi_{AB}\hat{n}_A\hat{n}_B + \frac{K_A}{2}\hat{n}_A^2 + \frac{K_B}{2}\hat{n}_B^2$ for $t_g = |\varphi|/(2\pi|\chi_{AB}|)$, and compute the average gate fidelity with the ideal CROT on the logical two-qubit subspace (leakage counted as error). At the strongest drive $t_g = 19 \mu\text{s}$, short compared with millisecond cavity lifetimes. Without self-Kerr the fidelity is unity to numerical precision, since $\hat{n}_A\hat{n}_B$ is the exact generator; with the simulated self-Kerr it lies between 0.07 and 0.43 over the whole drive range (Fig. 9, left), because a 5 kHz self-Kerr accumulates phases of order $\pi\bar{n}^2$ during the gate. A 99% CROT tolerates a residual self-Kerr of only $|K| \lesssim 7 \times 10^{-3}|\chi_{AB}|$, about 40 Hz for $\chi_{AB} = 5.8$ kHz (Fig. 9, right).

The original project explored one way out: a four-mode chain A – q_c – B – q_b with a second transmon and a second drive, so that the drive on q_c enhances χ_{AB} while the drive on q_b nulls K_B . The recovered data (Fig. 11) show such an operating point, with $\chi_{AB} \simeq 48$ kHz, $K_B = 0$, and a residual $K_A \simeq -12$ kHz. This is the side-transmon cancellation that the published scheme adopts.

D. The targeted resonance of Qin *et al.*

The project was completed after the above study by Qin, My, Copetudo, Kasper, Gao and Ng [13], whose resolution is not to drive harder but to drive at the right place. In the frame of a single-tone drive the dressed transmon can convert an a photon into a b photon plus one transmon excitation when $D^{\text{res}} = \omega_a - (\omega_b + \epsilon_{10}) \approx 0$. Near this resonance the cross-Kerr, which involves both cavities, is enhanced as $1/D^{\text{res}}$, whereas the self-Kerrs, which involve one cavity, are not; the residual self-Kerr of cavity a is then cancelled by a driven *side* transmon coupled only to a , the scheme already present in the four-mode chain of Fig. 11. Their parameters are $\omega_a/2\pi = 4.0$, $\omega_b/2\pi = 4.3$, $\omega_{10}/2\pi = 4.65$ GHz, $\alpha/2\pi = 0.2$ GHz, $|g_a/\delta_a| = 0.04$, $|g_b/\delta_b| = 0.03$ and $|\Omega_d/\delta_d| = 1$, with the drive detuning δ_d/α brought up to 0.835, just below the resonance at $\simeq 0.88$.

Feeding these parameters to the dressed-ladder code

reproduces the picture without any perturbation theory (Fig. 10). Approaching the resonance from below, $|\chi_{AB}|$ grows from 0.7 kHz to 37 kHz at the working point (on-off ratio $\simeq 52$; the perturbative estimate of Ref. [13] is > 60) and to 60 kHz just before the hybridisation window, while K_B decreases towards zero and K_A stays at a few kHz, so that $|\chi_{AB}/K_A| \simeq 4$ –9 instead of the $\simeq 1$ of the untargeted sweep. Inside the window $0.86 \lesssim \delta_d/\alpha \lesssim 0.91$ the dressed state $|1, 0, 0\rangle$ hybridises with $|0, 1, 1\rangle$ and the ladder is no longer defined; on the far flank, $\delta_d/\alpha \simeq 0.92$, the numerics even show K_A crossing zero with $\chi_{AB} \simeq 48$ kHz. For their error-correction example—an order-2 binomial data code, an order-1 cat ancilla and a first CROT angle $\varphi = 2\pi/(NM) = \pi$, gate time $14 \mu\text{s}$ at the working point—the two-mode average gate fidelity of our diagonal model is 0.43 with the central transmon alone and 0.69 once the side transmon cancels K_A ; but the remaining error is then a Kerr phase on the ancilla alone, which commutes with CROT and leaves the reduced state of the data mode untouched. The *data-rail* fidelity, which is the figure reported in Ref. [13], is 0.978 without and 1.000 with the side transmon in our model, consistent with their 0.96–0.99 (limited by higher-order and state-dependent Kerr terms that a diagonal model omits). With these gates they simulate the full Grimsom gadget and obtain an error-correction cycle of $\sim 19 \mu\text{s}$ that beats the bare cavity at high loss rates, at a cost of 7–12% relative to ideal CROT.

VI. CONCLUSION AND OUTLOOK

We followed a state from a test of quantumness to a bosonic code and back to the hardware. The spider state of Zaw *et al.* is a legitimate order- K rotation-symmetric primitive, and the only one whose preparation can be certified by sign measurements alone, through the spectral-gap witness of Eq. (8). As a code it detects $K - 1$ losses but does not correct them: its two codewords carry mean photon numbers in a fixed ratio of three, the Knill–Laflamme non-deformation condition fails by an amount of order $\bar{n}$, and any reweighting that repairs one condition abandons the maximal violation that defines the state while leaving its infinite energy intact. The spider is a witness, not a code.

The obstacle to rotation-code error correction is therefore in the gadgets, and we quantified the central one. An untargeted drive-tuned cross-Kerr interaction between two cavities gives a CROT gate in tens of microseconds, but the same mechanism produces self-Kerr of comparable size, and the gate tolerates only a residual of order 10^{-2} of the cross-Kerr. The published completion of the project [13] resolves this by placing a single-tone drive next to the $a \leftrightarrow b$ conversion resonance, which enhances the cross-Kerr by nearly two orders of magnitude while leaving the self-Kerrs small, and by cancelling the residual with a driven side transmon—the very chain explored in the recovered four-mode data. Our dressed-ladder nu-

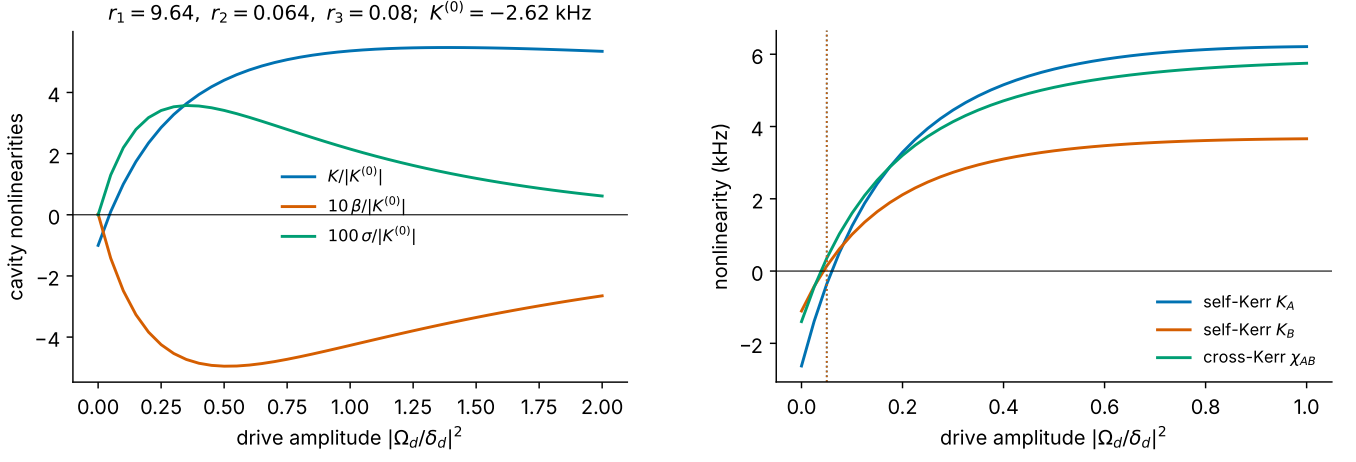


FIG. 8. Left: drive-induced nonlinearities of one cavity, normalised to the undriven $|K^{(0)}|$, in the format of Ref. [12]. Right: self-Kerr K_A , K_B and cross-Kerr χ_{AB} of two cavities sharing a driven transmon; dotted lines mark the Kerr-free points.

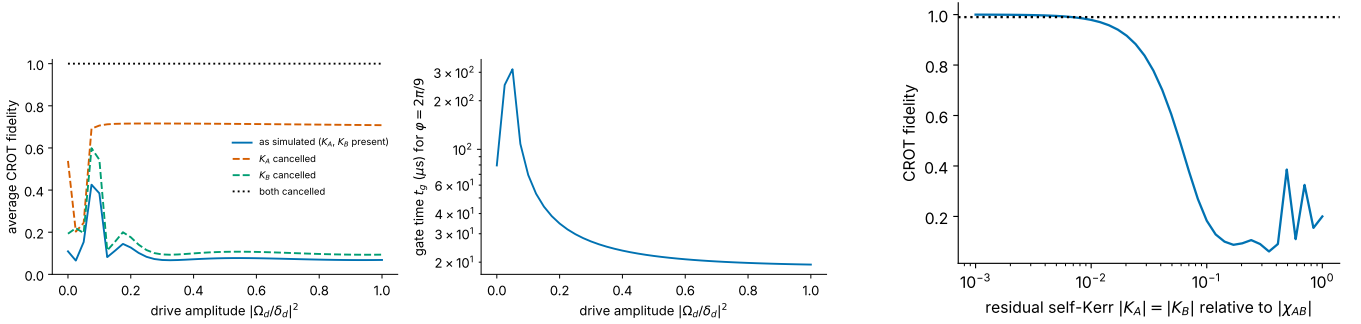


FIG. 9. Left: average fidelity of the cross-Kerr CROT ($\varphi = 2\pi/9$) as simulated and with one or both self-Kerrs cancelled, and the gate time, against the drive amplitude. Right: fidelity against the residual self-Kerr relative to the cross-Kerr; the dotted line is 0.99.

merics reproduce the resonance, the on-off ratio and the data-rail fidelity of that scheme.

Three directions follow. (i) *Beyond the diagonal model*: the 0.96–0.99 gate fidelities of Ref. [13] are limited by higher-order and state-dependent Kerr terms; a selective number-dependent phase (SNAP) correction or a Kerr echo could remove what the side transmon leaves. (ii) *Phase measurement*: the second half of the gadget needs a canonical phase measurement on the ancilla; the adaptive feedback scheme of Martin *et al.* [17] is the natural candidate, and the embedded phase uncertainty of Fig. 5 ranks the ancilla primitives for it. (iii) *Precession witness as a diagnostic*: although the spider code fails, the precession score is a cheap, device-independent non-classicality witness for *any* rotation-symmetric state, since $\hat{Q}_K$ is block diagonal in exactly the symmetry sectors the codes live in; using it to monitor cat- and binomial-code preparation seems the most useful legacy of the spider.

All numerics are reproducible from the accompanying repository, which contains the Python modules, the four notebooks that carry the argument of this paper, and the

recovered data and notes of the original project.

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Appendix A: Matrix elements of the half-line projector

With ψ_n the oscillator eigenfunctions, $-\frac{1}{2}\psi_n'' + \frac{1}{2}x^2\psi_n = (n + \frac{1}{2})\psi_n$. Multiplying the equations for m and n by ψ_n and ψ_m , subtracting and integrating over $x > 0$ gives $(m - n) \int_0^\infty \psi_m \psi_n dx =$

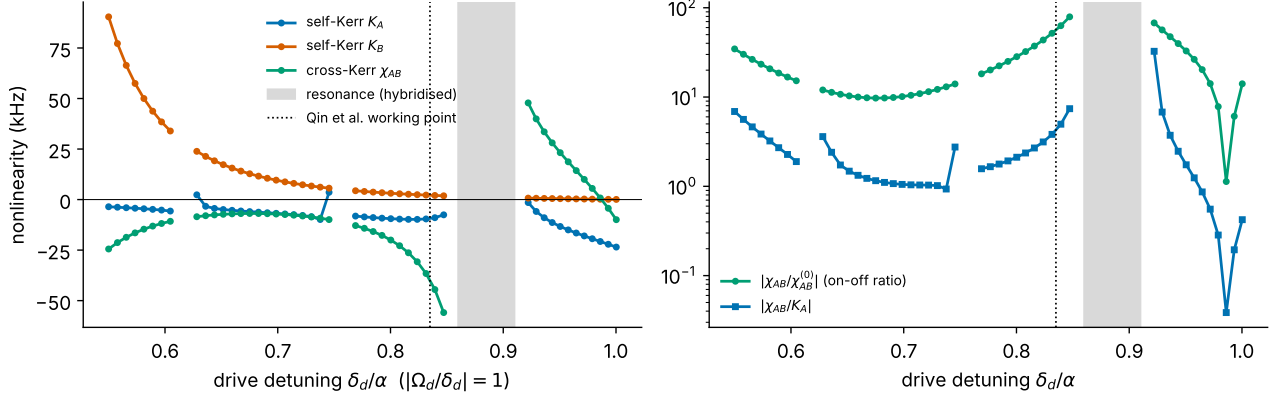


FIG. 10. Dressed-ladder reproduction of the targeted resonance of Qin *et al.* [13]. Left: K_A , K_B and χ_{AB} against the drive detuning at $|\Omega_d/\delta_d| = 1$; the shaded window is the hybridised region where the ladder is undefined, the dotted line their working point. Right: on-off ratio of the cross-Kerr and the ratio $|\chi_{AB}/K_A|$.

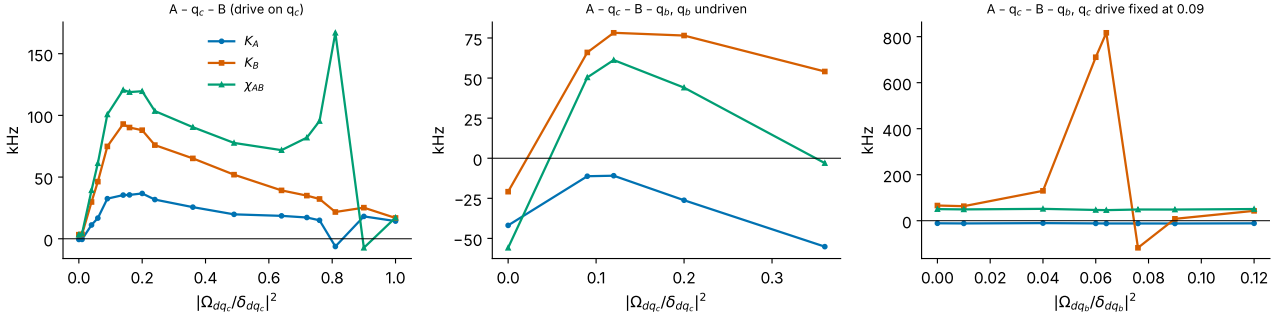


FIG. 11. Nonlinearities of the chains $A-q_c-B$ (left) and $A-q_c-B-q_b$ (middle, right) recovered from the original project ($\alpha/2\pi = 125$ MHz, larger couplings than in Fig. 8). In the right panel the q_c drive is fixed at $|\Omega/\delta|^2 = 0.09$ and the q_b drive is swept; K_B crosses zero near 0.074 while χ_{AB} stays near 50 kHz.

$\frac{1}{2} [\psi_n \psi'_m - \psi_m \psi'_n]_{x=0}$, which is the closed form used in the text. The boundary values are $\psi_{2j}(0) = \pi^{-1/4} (-1)^j \sqrt{(2j)!/(2^j j!)}$, $\psi_{2j+1}(0) = 0$, and $\psi'_n(0) = [\sqrt{n} \psi_{n-1}(0) - \sqrt{n+1} \psi_{n+1}(0)]/\sqrt{2}$, evaluated in log space for large n . The diagonal elements are 1/2 by parity.

Appendix B: Numerical details

The spider state is the top eigenvector of the residue-0 block of $\hat{Q}_K$, of dimension n for a Fock cutoff Kn ; dense diagonalisation up to $n = 7000$ takes under a minute. The Knill–Laflamme tensor is evaluated at a cutoff of 600 with the errors $\{\hat{1}, \hat{a}, \hat{a}^2, \hat{n}\}$. Cavity–transmon ladders use QuTiP operators with truncations (20, 8) for one cavity and (8, 8, 6) for two cavities; the smallest overlap between a coupled eigenstate and its bare label is 0.90 (one cavity) and 0.97 (two cavities). The CROT fidelity is $F = (|\text{tr} M|^2 + \text{tr} M^\dagger M)/[d(d+1)]$ with $d = 4$ and $M = B^\dagger U_{\text{ideal}}^\dagger U B$, B the orthonormal logical basis.

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